

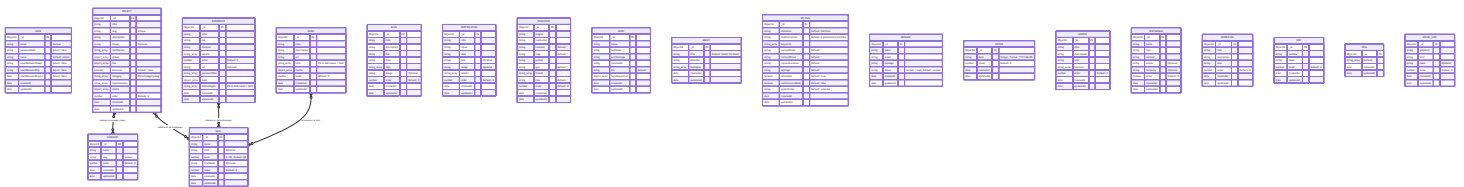
Database ER Diagram & Reference Manual

This document contains a visual Entity Relationship (ER) Diagram (using **Mermaid.js**) and a complete schema specification for the database of the portfolio application, structured via **Mongoose**.

Although MongoDB is a NoSQL, document-oriented database with largely denormalized collections, several key conceptual relationships connect the collections.

1. Visual Entity Relationship (ER) Diagram

Below is the conceptual ER diagram representing all **20 collections** and their logical relationships.



2. Collections and Schema Reference

2.1 Core Content Collections

1. Project (projects)

Detailed metadata representing custom software projects.

- **Fields:**
 - title (String, required): Name of the project.
 - slug (String, required, unique): Custom slug for URL path.
 - description (String, required): Long text detailing the project.
 - image (String, optional): Main teaser/preview image.
 - techNames (Array of Strings): Frameworks and tools used (conceptually mapping to Skill.name).

- `github` (Array of Objects): Source control repository configurations.
 - `label` (String, default: "Repository"): Display tag.
 - `url` (String, required): Git URL link.
- `live` (Array of Objects): Deployed link configurations.
 - `label` (String, default: "Live Demo"): Display tag.
 - `url` (String, required): Active deployment URL link.
- `featured` (Boolean, default: `false`): Highlights project on main landing showcases.
- `category` (Array of Strings): Conceptual many-to-many associations mapping to `Category.slug` .
- `improvements` (Array of Strings): Planned iterations or feature rollouts.
- `media` (Array of Objects): Showcase assets block.
 - `type` (String, enum: ["image", "video", "embed"] , default: "image"): Asset kind.
 - `url` (String, required): Media asset path/URL.
 - `caption` (String, optional): Explanation text overlay.
 - `thumbnail` (String, optional): Lazyload placeholder.
- `order` (Number, default: `0`): UI arrangement order.
- `createdAt` / `updatedAt` (Date, auto): Timestamps.

2. Category (`categories`)

Groups and filters portfolio items.

- **Fields:**

- `name` (String, required): Readable label (e.g. "Full-Stack Development").
- `slug` (String, required, unique): Router handle.
- `order` (Number, default: `0`): Render ranking.

3. Skill (`skills`)

Competency elements represented visually using charts/level meters.

- **Fields:**

- `name` (String, required): Label of the skill (e.g. "Next.js").
- `tech` (String, optional): Tech group classification.
- `level` (Number, default: `80`): Level between `0` and `100` .
- `iconName` (String, optional): Associated Lucide/React icon reference.
- `order` (Number, default: `0`): Drag order index.

4. Experience (`experiences`)

Timeline detailing job history and roles.

- **Fields:**

- `title` (String, required): Job title.
- `org` (String, required): Employing organization.
- `duration` (String, required): Employment duration span (e.g. "Jan 2025 - Present").
- `details` (Array of Strings): Highlights of duties and metrics achieved.
- `url` (String, optional): Organization landing page.
- `previousTitles` (Array of Strings): Other positions held during tenure.
- `links` (Array of Objects): External links related to this experience.
 - `label` (String, optional): Description label.
 - `url` (String, required): Target URL.
- `technologies` (Array of Strings): Tech stack used in this role.
- `order` (Number, default: `0`): Ordering sequence.

5. Education (`educations`)

Academic timeline record.

- **Fields:**

- `degree` (String, required): Level and field of study.
- `institution` (String, required): University/School name.
- `location` (String, default: `""`): Geography of school.
- `logo` (String, default: `""`): Institution logo file.
- `period` (String, required): Period of study.
- `gpa` (String, default: `""`): Academic GPA score.
- `details` (Array of Strings): Achievements, electives or distinctions.
- `link` (String, default: `""`): Institution portal URL.
- `order` (Number, default: `0`): Timeline order sequence index.

6. Certification (`certifications`)

List of professional courses and digital badges.

- **Fields:**

- `title` (String, required): Certification name.
- `issuer` (String, required): Awarding body.
- `date` (String, required): Time of issue.
- `link` (String, optional): Badge validation hyperlink.
- `image` (String, optional): Image/logo of the badge.
- `details` (Array of Strings): Included sub-topics or skills.
- `order` (Number, default: `0`): Sorting ordering field.

7. Demo (demos)

Playground or iframe-based live tools featured on the portfolio.

- **Fields:**

- `title` (String, required): Name of demo.
- `description` (String, required): Capabilities brief.
- `url` (String, required): Live target path.
- `tech` (Array of Strings): Main frameworks/languages.
- `media` (Array of Objects): Captioned image/video previews.
- `order` (Number, default: `0`): Interactive listing sequence.

8. Blog (blogs)

Self-hosted blog posts or external post links.

- **Fields:**

- `title` (String, required): Article title.
- `description` (String, required): Summary teaser.
- `link` (String, required): Article URL.
- `date` (String, required): Publishing date string.
- `tags` (Array of Strings): Keywords tag taxonomy.
- `image` (String, optional): Teaser cover graphic.
- `order` (Number, default: `0`): Listing rank index.

2.2 Configurations & Administration

9. User (users)

CMS panel administrator identity details.

- **Fields:**

- `email` (String, required, unique): Admin login email.
- `passwordHash` (String, required, select: `false`): Secure bcrypt hashed password.
- `name` (String, default: `"Admin"`): Visual profile name.
- `resetPasswordToken` (String, select: `false`): Reset token.
- `resetPasswordOtp` (String, select: `false`): Recovery OTP.
- `resetPasswordExpire` (Date, select: `false`): Recovery expiration.

10. Setting (settings)

Global layout toggles and metadata configurations.

- **Fields:**

- `siteName` (String, default: "Portfolio"): Visual tab bar name.
- `siteDescription` (String, default: "A professional portfolio"): Search engine meta description.
- `keywords` (Array of Strings): General keywords list.
- `contactEmail` (String, default: ""): Public contact address.
- `contactPhone` (String, default: ""): Public phone link.
- `contactLocation` (String, default: ""): City/State text.
- `ogImage` (String, default: ""): Open Graph metadata social share image.
- `isHireable` (Boolean, default: true): Status badge controller.
- `maintenanceMode` (Boolean, default: false): Standard lock-out shield toggle.
- `accentColor` (String, default: "emerald" , enum: ['emerald', 'blue', 'purple', 'rose', 'amber']): Accent color palette theme selector.

11. Hero (heroes)

Home page landing screen content values.

- **Fields:**

- `name / lastName` (String, required): First & Last name displays.
- `profileImage` (String): Hero headshot file.
- `resumeUrl` (String): Active resume PDF download link.
- `bio` (String, default: ""): Splash narrative subtitle.
- `typeSequences` (Array of Objects): Text loops with timing offsets:
 - `text` (String): Text string.
 - `delay` (Number, default: 2000): Hold time in milliseconds.

12. About (abouts)

Homepage detailed bio information.

- **Fields:**

- `title` (String, default: "Hello! I'm Shuvo"): Visual heading.
- `aboutBio` (String): Core about bio narrative.
- `highlights` (Array of Strings): Standout bullet achievements.

2.3 Site Utilities & Communications

13. Visitor (visitors)

Unique daily hit counts tracking records.

- **Fields:**
 - `date` (String, required, unique): Hits registration date string (YYYY-MM-DD).
 - `count` (Number, default: 0): Aggregated unique counters.

14. Message (messages)

Submissions generated via public contact forms.

- **Fields:**
 - `name` (String, required): Sender name.
 - `email` (String, required): Sender return email address.
 - `message` (String, required): Detailed body text.
 - `status` (String, default: "unread" , enum: ['unread', 'read']): Status flag for CMS control.

15. Service (services)

Custom professional service packages offered.

- **Fields:**
 - `title` (String, required): Service header label.
 - `description` (String, required): Service overview details.
 - `icon` (String, required): Lucide/React icon library map string.
 - `features` (Array of Strings): Key features/deliverables included.
 - `order` (Number, default: 0): UI render sorting value.

16. Testimonial (testimonials)

Colleague or client recommendations.

- **Fields:**
 - `name` (String, required): Reviewer's full name.
 - `role` (String, required): Reviewer's position.
 - `content` (String, required): Review body copy.
 - `avatar` (String, optional): Reviewer headshot path.
 - `company` (String, optional): Associated company/project name.
 - `order` (Number, default: 0): Layout sorting value.

17. Workflow (workflows)

Step-by-step dev process timeline nodes.

- **Fields:**
 - `title` (String, required): Workflow stage title (e.g. "Discover").
 - `description` (String, required): Detailed activities involved.
 - `icon` (String, required): UI icon label identifier.
 - `order` (Number, default: 0): Sequence rank.

18. Stat (stats)

Key resume highlights metrics.

- **Fields:**
 - `number` (String, required): Value string (e.g., "5+", "25+").
 - `label` (String, required): Metric label (e.g., "Open Source Contribs").
 - `order` (Number, default: 0): Layout order.

19. Tech (teches)

Simple static container cataloging active tech stacks.

- **Fields:**
 - `techList` (Array of Strings): Flat listing string catalog.

20. SocialLink (sociallinks)

External platform link configurations.

- **Fields:**
 - `platform` (String, required): Network name (e.g., "LinkedIn").
 - `href` (String, required): Profile link address.
 - `label` (String, optional): Hover text label tooltip details.
 - `order` (Number, default: 0): Ordering sequence.